Chapter 1

Reading KL Divergence as a Subtraction

Brando Miranda, October 2026. A textbook-style companion to the blog post *Reading KL Divergence as a Subtraction*. The figures are redrawn from my handwritten notes.

In this chapter. We build one grammar for the Kullback–Leibler divergence. Read $D_{\mathrm{KL}}(p\|q)$ as $p \overset{\mathrm{KL}}{-}_{p} q$: the divergence of p *from* q, measured on samples from p. Unrolled, it is literally a subtraction, $\mathbb{E}_{X\sim p}[\log p(X) - \log q(X)]$. The pointwise gap is antisymmetric like a vector, the average is not, and *all* of KL's asymmetry comes from which distribution we average under.

1.1 Why KL feels arbitrary

I have been doing machine learning since 2012, and KL divergence has shown up in my losses, my bounds, my RL regularizers and my variational objectives. Every time it wore a different costume:

- the **excess bits** paid for coding data from p with a code built for q;
- **cross-entropy minus entropy**, $H_p(q) - H_p(p)$, with minus signs to babysit;
- a **divergence**, which is "like a distance" until it isn't.

Each costume is correct. The trouble is having to switch mental models with each textbook. The goal of this chapter is to replace them with *one notation and one picture* from which every classical formula falls out. In an earlier post on score matching I subscripted the Fisher divergence with the distribution that carries the expectation, and promised to do the same for KL. Here that subscript turns out to carry the whole story of the asymmetry.

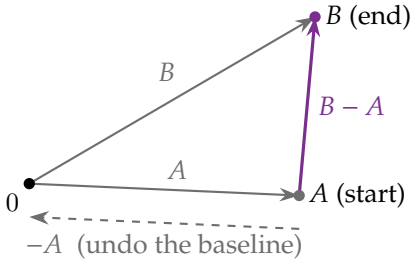
1.2 The displacement we already trust

Take two points A and B. The vector $B - A$ is the **displacement of** B **from** A: start at A, end at B. Whatever we subtract is the baseline, the place we start from. Its length is the ordinary distance $d(A, B) = \|B - A\|_2$. Figure 1.1(a) draws it the way I first sketched it: subtracting A means walking back along $-A$, then out along B.

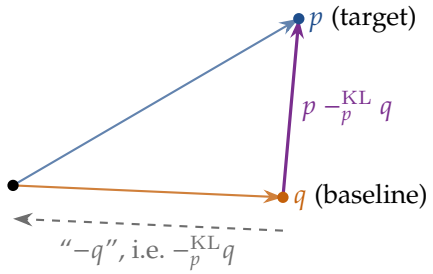
Everything we like about distances is visible in this picture. The length is nonnegative and zero only when $A = B$; it is symmetric; it obeys the triangle inequality. Keep one fact in mind for later:

> The **displacement is antisymmetric**, $A - B = -(B - A)$; the **length is symmetric**, because the norm forgets the sign.

That split is what explains KL.



(a) $B - A$: the displacement of B from A.



(b) The same grammar for distributions: q "goes to" p.

Figure 1.1: Displacement grammar, redrawn from page 1 of my notes. Panel (b) is the reading I propose for KL: $q \xrightarrow[p]{\mathrm{KL}} p$, start at the baseline q and arrive at the target p. On my first pass I had drawn the arrow the other way, $p \to q$; with the $B - A$ grammar it points from q to p.

1.3 Distributions as points, and the notation

Let p and q be distributions on the same space $\mathcal{X}$. On a finite space, $p \in [0,1]^{\mathcal{X}}$ is a vector with one coordinate per outcome x, its value $p(x)$, normalized so that $\sum_x p(x) = 1$. In the continuous case it is an "infinite vector" indexed by x, a density with $\int_{\mathcal{X}} p(x)\,dx = 1$. This is an analogy rather than a theorem, but it makes the right question obvious: *what is the displacement of p from q?*

Definition 1.1 (Displacement notation for KL)**.** For distributions p, q on $\mathcal{X}$ with $p \ll q$,

$$D_{\mathrm{KL}}(p\|q) \;=\; D_p^{\mathrm{KL}}(p\|q) \;=:\; p -_p^{\mathrm{KL}} q \;=\; \Delta_p^{\mathrm{KL}}(p,q) \;=\; \Delta_p^{\mathrm{KL}}(q \to p) \;=\; \Delta_p^{\mathrm{KL}}(p \leftarrow q).$$

Read it as **the divergence of p from q, measured on samples from p**. The superscript names the kind of subtraction; the subscript names the measure we average under, the "ruler". The baseline is q and the destination is p, exactly as $B - A$ has baseline A. Usually p is the target (the data) and q the model that approximates it.

"Divergence" here just means an unsymmetric distance, difference or displacement. Both arrows in Definition 1.1 say the same thing read in opposite directions: start at q, arrive at p. Figure 1.2 is the small spiral-pad version of the same idea, where the reading in bits sits next to the triangle.

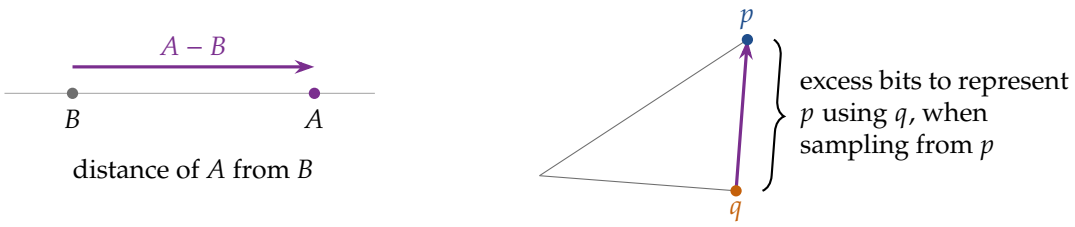
Put the two pictures side by side and the analogy reads

$$p -_p^{\mathrm{KL}} q \;=\; \mathbb{E}_{X\sim p}\big[\log p(X) - \log q(X)\big] \;\approx\; \text{"}A - B\text{"}, \tag{1.1}$$

the divergence of p from q next to the distance of A from B (Figure 1.3).

Remark 1.2 (A slight abuse of notation). The quotes around "$A - B$" matter. The vector $\vec{A} - \vec{B}$ is a displacement; the *distance* is the scalar $\|A - B\|_2$, and here "$A - B$" is shorthand for it. Likewise $p -_p^{\mathrm{KL}} q$ is not a vector: it is a scalar, like the Euclidean distance. Unlike that distance it is asymmetric, and the direction lives in the subscript.

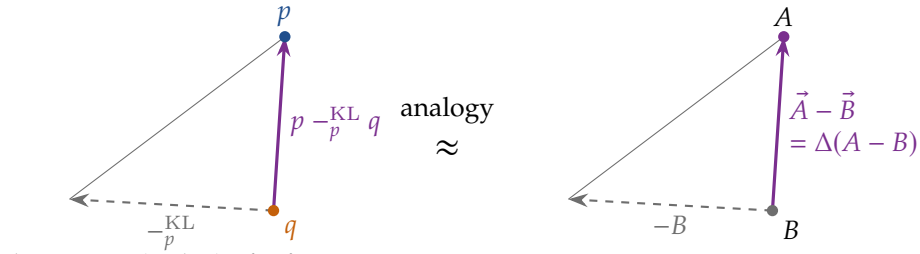
The same page asks a question in the margin: since we sample from p, does $\log p(X)$ come out the bigger side? **On average, yes**, by Gibbs' inequality (Proposition 1.4). **Not sample by sample**: a particular x drawn from p can still be likelier under q (Figure 1.5).



(a) On a line: $A - B$ is how far A sits from B.

(b) $p -_p^{\mathrm{KL}} q$, the "divergence" of p from q w.r.t. p.

Figure 1.2: Redrawn from page 6 of my notes. The one-dimensional picture (a) is the one everyone trusts. In (b) the vertical edge is the KL divergence and the brace gives its classical reading: $p^* -_{p^*}^{\mathrm{KL}} \hat{q} = H_{p^*}(\hat{q}) - H_{p^*}(p^*)$.



divergence (u-dist) of p from q
$= \Delta_p^{\mathrm{KL}}(q \to p)$

distance of A from B
"$A - B$" $\stackrel{?}{=} \|A - B\|_2$

Note: a little abuse of notation. $p -_p^{\mathrm{KL}} q$ is not a vector, it is a scalar like $\|A - B\|_2$. The Euclidean distance of A from B is symmetric; the divergence of p from q is not.

Figure 1.3: The vector view, redrawn from page 7 of my notes (its final version). Left: the divergence of p from q, with the measure p in the subtraction sign. Right: the displacement of A from B, with "$A - B$" standing for the distance.

1.4 Unroll it: KL really is a subtraction

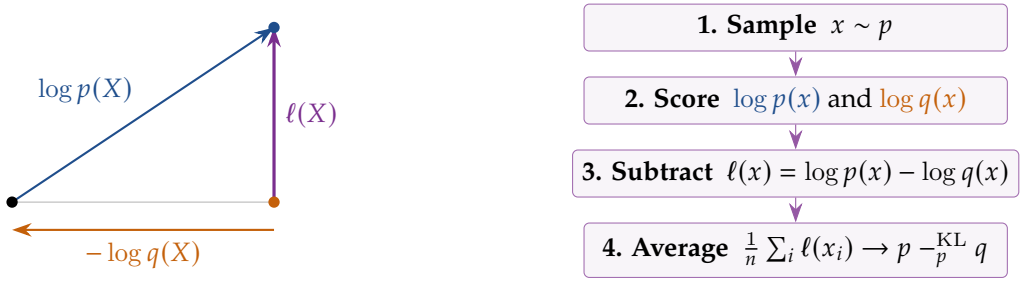
Write out the definition and the subtraction is sitting right there:

$$p -_p^{\mathrm{KL}} q \;=\; \mathbb{E}_{X\sim p}\big[\,\log p(X) \;-\; \log q(X)\,\big]. \tag{1.2}$$

Definition 1.3 (Log-likelihood displacement)**.** The pointwise log-likelihood displacement of p from q is $\ell(x) := \log p(x) - \log q(x)$. Then $p -_p^{\mathrm{KL}} q = \mathbb{E}_p[\ell(X)]$.

Read (1.2) as a procedure (Figure 1.4b): sample $x \sim p$; score it under both models; subtract, which tells you how much likelier p makes this x than q does, in log units (nats with $\ln$, bits with $\log_2$); average. So $D_{\mathrm{KL}}(p\|q)$ is **the average log-likelihood advantage of p over q on p's own data**. No codes, and no minus signs to keep track of.

Proposition 1.4 (Gibbs' inequality)**.** $p -_p^{\mathrm{KL}} q \geq 0$, *with equality if and only if $p = q$ almost everywhere.*



(a) The triangle in log-likelihood space. **(b)** KL as a recipe.

Figure 1.4: Redrawn from page 5 of my notes, the page where it clicked. (a) Walking out along $\log p(X)$ and back along $-\log q(X)$ leaves the displacement $\ell(X) = \log p(X) - \log q(X)$, just as $B - A = B + (-A)$. (b) Sample, compute log-probabilities, compare, average.

Proof. By Jensen's inequality for the concave log,

$$-\mathbb{E}_p[\ell(X)] = \mathbb{E}_p\left[\log \frac{q(X)}{p(X)}\right] \le \log \mathbb{E}_p\left[\frac{q(X)}{p(X)}\right] = \log \int_{\{p>0\}} q(x)\,dx \le \log 1 = 0,$$

with equality only when q/p is constant on the support of p, that is, when $p = q$. □

For a particular x, $\ell(x)$ can be negative: q may like some of p's samples more than p does. On average, on p's own turf, p always wins or ties. Figure 1.5 shows both facts at once.

Example 1.5 (Sometimes it literally is a distance)**.** Let $p = \mathcal{N}(\mu_p, \sigma^2)$ and $q = \mathcal{N}(\mu_q, \sigma^2)$. Then $\ell(x) = \frac{(x-\mu_q)^2 - (x-\mu_p)^2}{2\sigma^2}$, and averaging under p gives

$$p \underset{p}{-^{\mathrm{KL}}} q = \frac{(\mu_p - \mu_q)^2}{2\sigma^2},$$

half the squared Euclidean distance between the means, in units of σ (Figure 1.6). In the simplest case the displacement picture is not a mnemonic; it is exactly the old distance.

1.5 Where the asymmetry lives

Swap p and q in the pointwise displacement and it flips sign, exactly like a vector: $\log q(x) - \log p(x) = -\ell(x)$. So the reverse divergence is

$$q \underset{q}{-^{\mathrm{KL}}} p \;=\; \mathbb{E}_{X\sim q}\big[-\ell(X)\big]. \tag{1.3}$$

Where the asymmetry comes from

Same pointwise displacement, opposite sign, **different averaging distribution**. In Euclidean space the norm forgets the sign and symmetry comes for free. In KL the "norm" is an average taken under one of the two endpoints; change the endpoint you sample from and you change the number. The subscript in $\underset{p}{-^{\mathrm{KL}}}$ records exactly this.

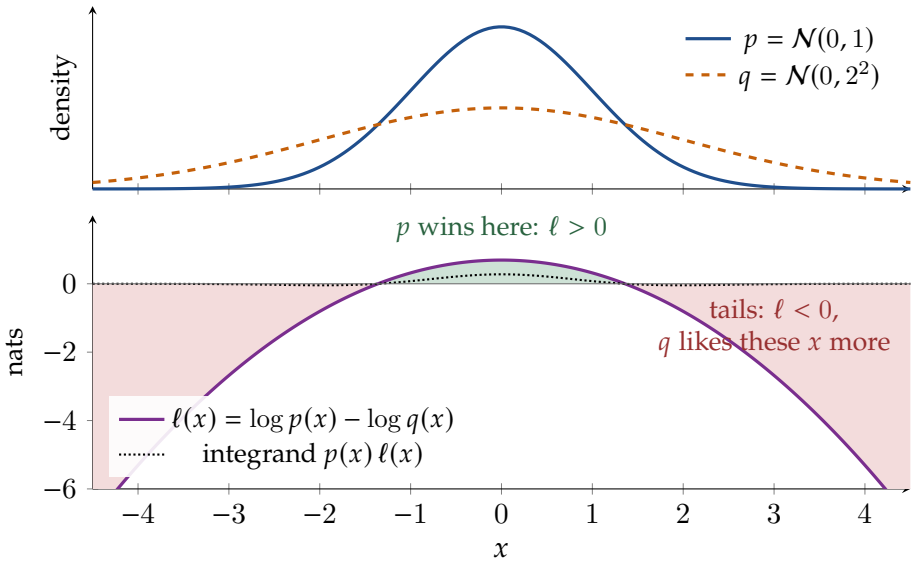


Figure 1.5: Pointwise versus average, for $p = \mathcal{N}(0,1)$ and $q = \mathcal{N}(0,2^2)$, where $\ell(x) = \ln 2 - \frac{3}{8}x^2$. Samples from p with $|x| > \sqrt{(8\ln 2)/3} \approx 1.36$ (about 17% of them) are *likelier* under q. The dotted integrand still has positive total area, $p\,\mathrm{KL}_p\,q = \ln 2 - \frac{3}{8} \approx 0.32$ nats, as Gibbs' inequality promises.

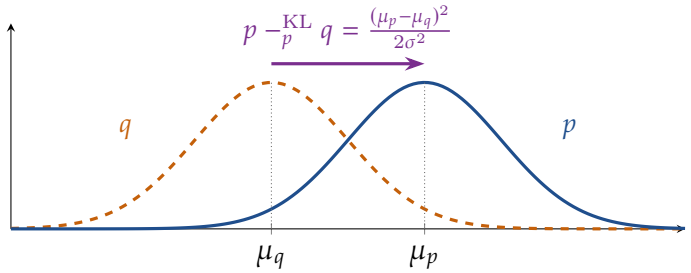


Figure 1.6: Example 1.5: with equal variances the KL divergence is the displacement of the means from baseline μ_q to target μ_p, squared and halved ($\sigma = 1$ and $\mu_p - \mu_q = 2$ give 2 nats). Here it is symmetric; the asymmetry appears as soon as the shapes differ.

Example 1.6 (Narrow versus wide)**.** Let $p = \mathcal{N}(0,1)$ and $q = \mathcal{N}(0,2^2)$. Then

$$p\,\mathrm{KL}_p\,q = \ln 2 - \tfrac{3}{8} \approx 0.32 \text{ nats}, \qquad q\,\mathrm{KL}_q\,p = \tfrac{3}{2} - \ln 2 \approx 0.81 \text{ nats}.$$

On p's data the wide q is a little diffuse but never shocked, so the cost is low. On q's data the narrow p keeps meeting samples in the tails that it thought nearly impossible, so the cost is high.

This is the fact behind "forward KL is mode-covering, reverse KL is mode-seeking", and behind the different behaviour of maximum likelihood (forward) and variational inference (reverse). Fit a single Gaussian q to a two-mode p in each direction and the subscript decides everything (Figure 1.7).

Koller and Friedman's treatment of M- and I-projections [9, Sec. 8.5] is the cleanest account

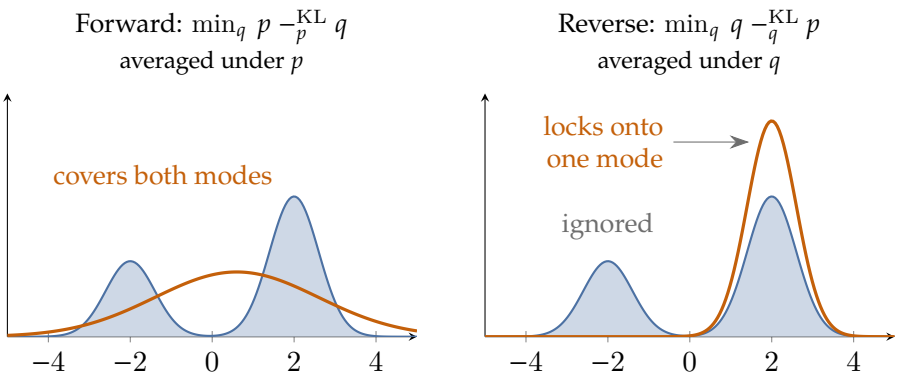


Figure 1.7: The best single Gaussian q (orange) for the two-mode $p = 0.35\,\mathcal{N}(-2, 0.6^2) + 0.65\,\mathcal{N}(2, 0.6^2)$ (shaded). Left: averaged under p, any place where p has mass and q does not costs a huge $\ell(x)$, so q stretches to cover everything; the optimum matches moments, $q = \mathcal{N}(0.6, 2.0^2)$, at 0.56 nats. Right: averaged under q, q is scored only where it puts its own samples, so it may ignore a whole mode; the optimum is $q \approx \mathcal{N}(2.0, 0.6^2)$, at $0.43 \approx -\ln 0.65$ nats.

of this I know. If a symmetric quantity is needed, add the two directions (Jeffreys' divergence):

$$p\ \mathrm{KL}_p\ q\ +\ q\ \mathrm{KL}_q\ p\ =\ \int \big(p(x) - q(x)\big)\big(\log p(x) - \log q(x)\big)\,dx. \tag{1.4}$$

A difference in probability space times a difference in log space: both subtractions in one integrand.

1.6 How KL is, and is not, like a distance

Table 1.1 sets KL beside the quantity we started from.

	Half squared Euclidean, $\frac{1}{2}\|B - A\|^2$	KL, $p\ \mathrm{KL}_p\ q$
Pointwise displacement	$B - A$	$\log p(x) - \log q(x)$
Swap the endpoints	flips the displacement's sign	flips the displacement's sign
Nonnegative, zero only at equality	yes	yes (Gibbs)
Symmetric	yes	no: the average is under one endpoint
Triangle inequality	no (the unsquared distance has it)	no
Bregman divergence of	$\frac{1}{2}\|x\|^2$	negative entropy $\sum_x p(x) \log p(x)$
Locally, near equality	itself	$\frac{1}{2}\,\delta^\top F(\theta)\,\delta$, F the Fisher information

Table 1.1: KL next to half the squared Euclidean distance.

The honest summary: **KL is not a distance, but it behaves like half a squared distance.** Both belong to the family of Bregman divergences,

$$D_F(p, q)\ =\ F(p) - F(q) - \langle \nabla F(q),\ p - q \rangle,$$

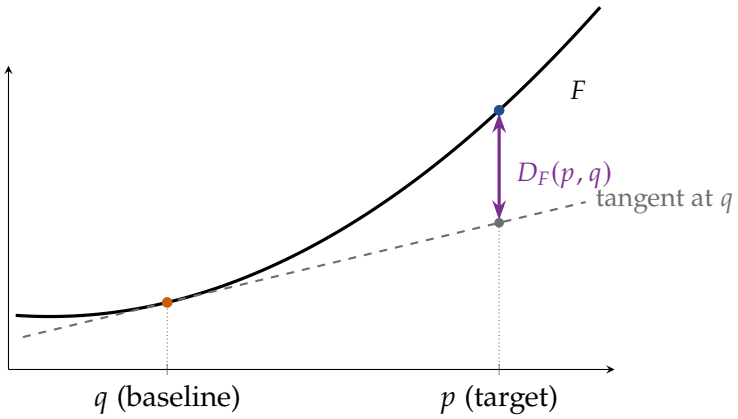


Figure 1.8: A Bregman divergence is the gap at the target p between a convex F and its tangent at the baseline q. Moving the tangent point changes the gap, which is why such divergences are asymmetric in general. Half the squared Euclidean distance ($F = \frac{1}{2}\|x\|^2$) and KL (F = negative entropy) are both of this form.

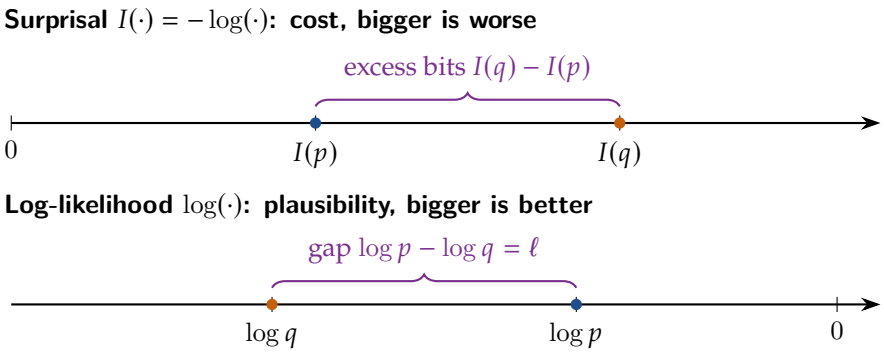


Figure 1.9: Redrawn from page 4 of my notes, the margin of a book chapter (p the reference distribution we sample from, q the approximating one). On the surprisal axis KL is $\mathbb{E}_p[I(q) - I(p)] = H_p(q) - H_p(p)$; flip the axis and the same gap is $\mathbb{E}_p[\log p - \log q]$. Drawn for a typical $x \sim p$, where p wins.

the gap at p between a convex function F and its tangent drawn at the baseline q (Figure 1.8). With $F(x) = \frac{1}{2}\|x\|^2$ this is $\frac{1}{2}\|p - q\|^2$; with the negative entropy it is $D_{\mathrm{KL}}(p\|q)$. Nearby distributions see a quadratic form, $D_{\mathrm{KL}}(p_\theta\|p_{\theta+\delta}) \approx \frac{1}{2}\delta^\top F(\theta)\delta$, so locally KL is a squared length in the Fisher metric. It even has a Pythagorean identity for projections onto nice families [4]. What it gives up is symmetry, and we now know exactly why.

1.7 Why I would teach the log-likelihood view first

The signs make sense. The bits view counts *cost*, so it is built from $-\log$ terms, and KL becomes cost minus cost with sign flips to babysit. The log-likelihood view counts *plausibility*: bigger is better, and KL is the plausibility gap. It is the same gap on a mirrored axis (Figure 1.9).

Maximum likelihood falls out in one line. With p the data distribution and q_θ the model,

$$p \underset{p}{-}^{\mathrm{KL}} q_\theta \;=\; \underbrace{\mathbb{E}_p[\log p(X)]}_{\text{independent of } \theta} \;-\; \mathbb{E}_p[\log q_\theta(X)].$$

Minimizing over θ is maximizing the expected log-likelihood, and with data we replace $\mathbb{E}_p$ by the empirical average. **Maximum likelihood minimizes the forward KL from the data to the model**: we move the baseline q_θ toward the data, scored on the data's own samples.

It connects to the rest of statistics. $\ell(x)$ is the log-likelihood ratio, the central object of hypothesis testing, Bayesian evidence, importance sampling and policy-gradient RL. Once KL is "average ℓ", much of ML stops looking like a collection of unrelated tricks (Appendix A).

1.8 Notes and history

I did not get this view from a course or a textbook. I got it while working through these notes with an AI model: I asked about the excess-bits formula, and it rewrote KL as $\mathbb{E}_p[\log p - \log q]$ without my asking.

It is not new as mathematics. Kullback and Leibler [11] introduced the quantity as the mean information for discrimination between two hypotheses per observation from one of them, which is precisely the expected log-likelihood ratio, and Kullback [10] preferred the name *discrimination information*. Akaike built AIC on the same reading [1], and Etz [5] calls KL "the expected log likelihood ratio in favor of the true distribution." What I could not find in the books on my shelf, those I was assigned or recommended at UIUC, MIT and Stanford, is any of them *leading* with it as the intuition (Table 1.2). Nor could I find it folded into one displacement grammar in which the averaging distribution is part of the notation and explains the asymmetry. That part is my framing.

Why did it take fourteen years? My guess, with no complaint against anyone, is that a class, a talk or a paper has a syllabus or a page limit, and "why is it a ratio, and why that sign?" feels like something one should already know. So nobody stops to rebuild the foundations. AI changes the economics of asking: a tireless sparring partner that never makes you feel slow. A curious person who works hard with it can now understand the foundations more deeply than any course gave them.

Exercises

Exercise 1.1. Derive Example 1.5: for $p = \mathcal{N}(\mu_p, \sigma^2)$ and $q = \mathcal{N}(\mu_q, \sigma^2)$, show $p \underset{p}{-}^{\mathrm{KL}} q = (\mu_p - \mu_q)^2/(2\sigma^2)$ by averaging $\ell(x)$ under p.

Exercise 1.2. For $p = \mathcal{N}(0, 1)$ and $q = \mathcal{N}(0, s^2)$, compute $p \underset{p}{-}^{\mathrm{KL}} q$ and $q \underset{q}{-}^{\mathrm{KL}} p$ as functions of s. For which s are they equal? Check the values 0.32 and 0.81 at $s = 2$.

Exercise 1.3. Prove the Jeffreys identity (1.4), and show its integrand is pointwise nonnegative. Why does this not make each direction nonnegative pointwise?

Exercise 1.4. Let $y = g(x)$ with g smooth and invertible, and let p_Y, q_Y be the induced densities. Show that ℓ is unchanged, $\log p_Y(y) - \log q_Y(y) = \log p(x) - \log q(x)$, so $p \underset{p}{-}^{\mathrm{KL}} q$ is the same in both coordinates, while the pointwise difference $p_Y(y) - q_Y(y)$ is rescaled by the Jacobian.

Book	Expected log-difference or log-ratio?	Sampled log-likelihood form?	Headline intuition	Where
Hogg, McKean & Craig [8]	Yes, unnamed: $\mathbb{E}_{\theta_0}\big[\log \frac{f(X;\theta)}{f(X;\theta_0)}\big]$	Yes: average log-likelihood ratio over the sample	None; KL never named	Thm. 6.1.1 and proof
MacKay [12]	Yes, $\sum_x P(x) \log \frac{P(x)}{Q(x)}$	Partly: expected weight of evidence, not linked to KL	Wasted bits	Sec. 2.6; ch. 5; Secs. 18.3–18.4
Hastie, Tibshirani & Friedman [7]	Only as ICA's mutual information	No	Dependence	Sec. 14.7.2
Koller & Friedman [9]	Yes; a proof writes $\mathbb{E}_P[\log P - \log P']$	Yes	Extra coding cost	Def. A.5; Prop. 16.1; Sec. 17.2.5
Shalev-Shwartz & Ben-David [13]	Yes, inside the log-loss risk	Yes: MLE as ERM with log-loss	Log-loss risk = relative entropy + entropy	Sec. 24.1
Goodfellow, Bengio & Courville [6]	Yes, exactly $\mathbb{E}_{x\sim P}[\log P(x) - \log Q(x)]$	Yes: MLE as KL to the empirical distribution	Extra bits	Eq. 3.50; eq. 5.60
Bishop & Bishop [3]	Cross-entropy minus entropy, then $-\int p \ln \frac{q}{p}$	In Ex. 2.34	Extra information from the wrong code	Sec. 2.5.5
Bishop [2]	Cross-entropy minus entropy, then $-\int p \ln \frac{q}{p}$	Yes: $\frac{1}{N}\sum_n \{-\ln q(x_n\|\theta) + \ln p(x_n)\}$	Extra information to transmit	Sec. 1.6.1

Table 1.2: How the books on my shelf introduce KL. The subtraction is in almost all of them; the headline is always bits.

Exercise 1.5. Show that the Gaussian q minimizing $p \,-\!\!\!

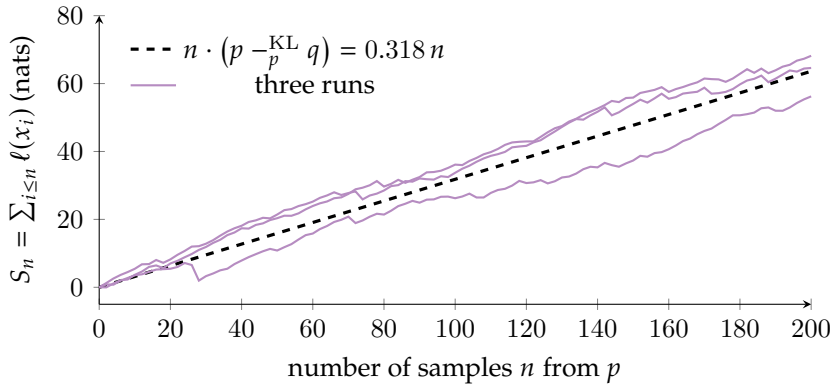


Figure A.1: Evidence adds. Cumulative log-likelihood ratio for $p = \mathcal{N}(0, 1)$ against $q = \mathcal{N}(0, 2^2)$ on samples from p. Single steps can be negative (Figure 1.5), but the sum drifts up at slope $p -_p^{\mathrm{KL}} q \approx 0.32$ nats per sample. Paths generated by `evidence_paths.py`.

about distributions, not coordinates.

- **Bayes is additive in log-ratios.** Posterior odds are prior odds times the likelihood ratio, so $\ell(x)$ is how far one observation moves your log-odds toward p, Good and Turing's *weight of evidence*.
- **Hypothesis testing.** By the Neyman–Pearson lemma, thresholding the likelihood ratio is the most powerful test of p against q. By Stein's lemma, with n samples from p the best achievable probability of wrongly accepting q decays like $e^{-nD_{\mathrm{KL}}(p\|q)}$: KL is the exponent of distinguishability.
- **Changing measure needs a ratio.** Importance sampling uses $\mathbb{E}_p[f(X)] = \mathbb{E}_q\left[\frac{p(X)}{q(X)} f(X)\right]$. That is PPO's ratio, **new over old**, reweighting samples collected by the old policy. Averaged under the new policy, $\log r$ gives $D_{\mathrm{KL}}(\pi_\theta \| \pi_{\mathrm{old}})$; on the old policy's samples, $\mathbb{E}_{\mathrm{old}}[-\log r]$ gives $D_{\mathrm{KL}}(\pi_{\mathrm{old}} \| \pi_\theta)$, the direction TRPO constrains.

Differences of probabilities do give real distances, such as total variation $\frac{1}{2}\sum_x |p(x) - q(x)|$, and Pinsker's inequality bounds it by KL: $\mathrm{TV}(p, q) \le \sqrt{D_{\mathrm{KL}}(p\|q)/2}$. They just lack the properties above, which is why ratios keep appearing everywhere in ML.

B The classical bits view, side by side

The textbook route starts from surprisal. Seeing x with probability $p(x)$ costs $-\log_2 p(x)$ bits; for example $p(x) = 2^{-3}$ costs 3 bits (Figure B.1). Averaging gives entropy. With the sampling distribution as a subscript, as everywhere in this chapter,

$$H_p(q) := \mathbb{E}_{X\sim p}[-\log q(X)] \quad \text{(cross-entropy)}, \qquad H_p(p) = \mathbb{E}_{X\sim p}[-\log p(X)] \quad \text{(entropy)}.$$

$H_p(q)$ is the expected code length when the data come from p but the code was built for q; $H_p(p)$ is the best possible. Their difference is the classical KL, the **excess bits** of using the wrong code:

$$p -_p^{\mathrm{KL}} q \;=\; H_p(q) - H_p(p) \;=\; \mathbb{E}_p[-\log q(X)] - \mathbb{E}_p[-\log p(X)] \;=\; \mathbb{E}_p[\log p(X) - \log q(X)].$$

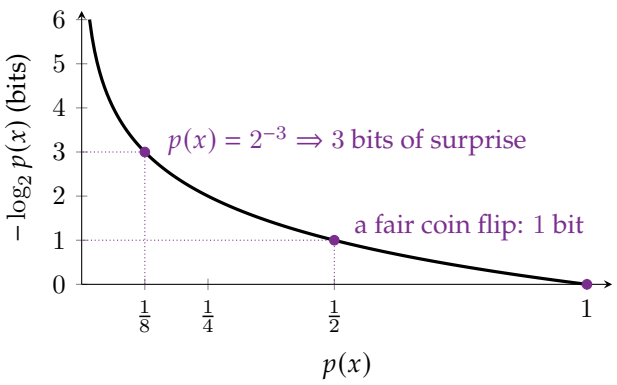


Figure B.1: Surprisal, from page 5 of my notes: a bit of information is $-\log_2 p(x)$ when $X = x$. Rare outcomes cost more bits; a sure outcome costs none.

The two readings are the same number thought of two ways.

- **Bits:** a difference of *costs*. Bigger is worse, so every term carries a minus sign, and we subtract the optimal cost from the suboptimal one.
- **Log-likelihood:** a difference of *plausibilities*. Bigger is better, there are no minus signs, and we subtract the baseline model's log-likelihood from the true model's, on the true model's data.

The bits view is the right one when compression is the goal. For learning, where everything we optimize is already a log-likelihood, the second is far easier to hold in one's head. They are the same subtraction.

Bibliography

[1] H. Akaike. Information theory and an extension of the maximum likelihood principle. *2nd International Symposium on Information Theory*, 1973.

[2] C. M. Bishop. *Pattern Recognition and Machine Learning*. Springer, 2006.

[3] C. M. Bishop and H. Bishop. *Deep Learning: Foundations and Concepts*. Springer, 2024.

[4] I. Csiszár. I-divergence geometry of probability distributions and minimization problems. *Annals of Probability*, 3(1):146–158, 1975.

[5] A. Etz. Some technical notes on Kullback–Leibler divergence. Blog post, 2018.

[6] I. Goodfellow, Y. Bengio and A. Courville. *Deep Learning*. MIT Press, 2016.

[7] T. Hastie, R. Tibshirani and J. Friedman. *The Elements of Statistical Learning*, 2nd ed. Springer, 2009.

[8] R. V. Hogg, J. W. McKean and A. T. Craig. *Introduction to Mathematical Statistics*, 7th ed. Pearson, 2013.

[9] D. Koller and N. Friedman. *Probabilistic Graphical Models*. MIT Press, 2009.

[10] S. Kullback. *Information Theory and Statistics*. Wiley, 1959.

[11] S. Kullback and R. A. Leibler. On information and sufficiency. *Annals of Mathematical Statistics*, 22(1):79–86, 1951.

[12] D. J. C. MacKay. *Information Theory, Inference, and Learning Algorithms*. Cambridge University Press, 2003.

[13] S. Shalev-Shwartz and S. Ben-David. *Understanding Machine Learning*. Cambridge University Press, 2014.